

Name: _____

Period: _____

Date: _____

Unit 3 Exam Review**Simplifying Exponentials and Logarithms**

Simplify each expression (evaluate if possible). If the expression does not exist, write “DNE.”

1) $4^{5/2}$	2) $216^{2/3}$	3) $49^{3/2}$	4) $125^{4/3}$	5) $27^{3/2}$
6) $\log 1000$	7) $\log_3 9$	8) $\log_2 32$	9) $\log_2(0.25)$	10) $\log_5 \frac{1}{5}$
11) $\ln 1$	12) $\ln e^7$	13) $\ln 0$	14) $\ln \frac{1}{e}$	15) $\ln(-e)$
16) $\left(\frac{16}{9}\right)^{-\frac{1}{2}}$	17) $\left(\frac{2x^3y^{-1}}{z^{-\frac{1}{2}}}\right)^6$	18) $\left(\frac{a^{-2}b^3}{8a^{10}b^0}\right)^{-\frac{1}{3}}$	19) $(16x^{-21}y^{33})^{-2/3} \cdot (100x^{12}y^{-4})^{1/2}$	
20) $10^{\frac{1}{2} \log 81}$	21) $e^{-2 \ln 4}$	22) $-2 \log_7 \left(\frac{1}{49}\right)$	23) $\log_3 \sqrt{27}$	24) $\log_6 \left(\frac{1}{\sqrt[3]{36}}\right)$

Expand and Condense Logarithms

For #'s 25-27, condense the expression into a single logarithm.

For #'s 28-30, expand the logarithmic expression (simplify each logarithm).

25) $2 \log x - 3 \log y - 5 \log z$	26) $-\frac{1}{2} \ln(a) + \frac{2}{3} \ln(b) + \frac{3}{5} \ln(c)$	27) $3 \log_4(x) - 3$
28) $\log_2 \left(\frac{4x^5 \cdot \sqrt{y}}{\sqrt[3]{z^2}}\right)$	29) $\ln \left(\frac{\sqrt[12]{a^7}}{e^7}\right)$	30) $\log_3(18\sqrt{xy})$

Exponentials and Logarithms in base eFor #'s 31-34, solve for x . Write your answer in calculator-ready form.

*Note: you might want to practice expressing your answer as a decimal OR a fraction – the exam could use either format.

*For #32, see if you can write the answer using only fractions and no negative coefficient(s).

*Unlikely you will have a problem like #34, but it is really great practice – ESPECIALLY for what's coming.

31) $15 + 42e^{-0.25x} = 50$	32) $\frac{500}{1+30e^{-1.5x}} = 20$
33) $\ln(2x + 3) - \ln x = 2$	34) $\log_3(x - 4) + \log_3(x + 2) = 3$
35) Mr. Spake invested \$1500 in an account with an interest rate of 4.3% compounded continuously. How many years will it take for Mr. Spake's account to accrue \$7500? Write your answer in calculator-ready form. *Note: To accrue \$7500 means Mr. Spake gained \$7500 from his investment, so what would the account's <i>total</i> be? This vocabulary won't be on the test but Mr. Spake wants you to understand the meaning of the word “accrue” ☺	

Exponential Application Problems

For #'s 36-39, write the exponential equation modeling the given scenario. If solving for an unknown variable, write your answer in calculator-ready form.

36) Herbie has \$35,000 to invest (one could say that Herbie is “fully loaded”). Herbie’s bank offers a savings account that has an APR of 1.75% and compounds monthly. Write an equation that Herbie could use to know how much money he will have in the account after 6 years.
37) Wrex opened a savings account with Citadel Central Bank. His account has an APR of 2.25% compounded quarterly. If Wrex opens his account with \$3000, how long will it take for the account to earn \$12000?
38) A sample of Compound- Γ (read: “compound gamma”) originally contained 80 grams. After 50 days only 2.5 grams of the original sample remain. What is the half life of Compound- Γ ?
39) The half-life of Mr. Spake’s cortisol levels when he is writing test reviews for his OnRamps Precalculus class is 45 minutes. Mr. Spake’s cortisol level was 12 mcg/dL when he first started writing this test review. How long will it take for his cortisol levels to reduce by 70%? (Just FYI this is <i>not</i> realistic).

Graphs of Exponential and Logarithmic Functions

For #'s 40-42, draw a sketch for each logarithmic function, name its transformations, and give its domain and range. For #'s 43-45, identify the function as growth/decay, find the y- and x-intercepts, and give the equation of its asymptote.

40) $f(x) = 3 \log_2(-x) + 1$	41) $f(x) = -\frac{1}{2} \ln(x + 4)$	42) $f(x) = 4 - \log(2x - 2)$
43) $f(x) = 3 \left(\frac{5}{2}\right)^x + 2$	44) $f(x) = 8 - 4 \left(\frac{2}{3}\right)^x$	45) $f(x) = \frac{1}{2} (7)^{-x} - 1$

Writing Exponential Models

46) Determine the equation of the exponential function in the form $f(x) = a \cdot b^x$ that passes through the following points: $\left(-1, \frac{1}{12}\right)$ and $\left(2, \frac{9}{4}\right)$.	47) The table models an exponential situation; create a model in the form $y = a \cdot b^t$. <table border="1" data-bbox="1068 1314 1268 1423"><tr><th>t</th><th>y</th></tr><tr><td>-2</td><td>28</td></tr><tr><td>2</td><td>16/3</td></tr></table>	t	y	-2	28	2	16/3
t	y						
-2	28						
2	16/3						
48) A house is purchased for \$450,000 and its value appreciates by 3.75% every 4 years. a) Write an exponential model: $V(t) = a \cdot b^{kt}$. b) How many years will it take for the house to double its value? Solve for t using natural logarithms (ln).	49) A new disease, Skibidi-Toilet-Rizz-Ohio-Kappa-Extra, abbreviated S.T.R.O.K.E., infects students with brain rot. The students affected by brain rot is modeled by the logistic function $B(t) = \frac{200}{5 + 20e^{-0.3x}}$ a) How many students are initially infected with brain rot? b) How many days will it take for 30 students to have S.T.R.O.K.E.?						